

# Personalized Treatment Planning Using Patient History Data

## 1. Background & Context

Healthcare systems generate large volumes of patient data including demographics, medical history, lifestyle behaviors, treatments, and clinical outcomes. Despite this availability, many treatment decisions remain standardized. This project applies data analytics and machine learning to support personalized treatment planning.

## 2. Problem Statement

The goal is to analyze historical patient data and build predictive models that assist healthcare providers in recommending personalized treatments, estimating recovery outcomes, and identifying relapse risk.

## 3. Project Objectives

- Perform exploratory data analysis to understand patient demographics and clinical patterns.
- Identify lifestyle and medical factors influencing treatment outcomes.
- Engineer meaningful features representing patient risk and treatment effectiveness.
- Develop machine learning models for outcome prediction and patient segmentation.

## 4. Dataset Overview

The dataset consists of 9,000 synthetic yet realistic patient records. Each record represents a single treatment episode and includes demographics, medical history, lifestyle factors, treatment details, and clinical outcomes.

## 5. Exploratory Data Analysis (EDA)

EDA involves examining data distributions, correlations, and patterns. Analysis includes demographic profiles, disease prevalence, lifestyle impact on recovery, treatment effectiveness, and relapse trends.

## 6. Feature Engineering

New features are derived to improve model performance, such as lifestyle risk scores, chronic condition counts, treatment intensity measures, and composite patient risk indices.

## 7. Machine Learning Approach

Unsupervised learning techniques are used to segment patients into clinically meaningful groups. Supervised learning models are developed to predict relapse probability, recovery time, and improvement scores.

## 8. Model Evaluation

Models are evaluated using appropriate metrics including precision, recall, ROC-AUC, and regression error measures. Model explainability is emphasized to ensure clinical relevance.

## 9. Ethical & Compliance Considerations

The project emphasizes responsible data usage, bias detection, model transparency, and human-in-the-loop decision support to align with ethical healthcare analytics practices.

## 10. Final Deliverables

- EDA report with insights and visualizations
- Feature-engineered dataset
- Trained and evaluated machine learning models
- Data-driven treatment planning recommendations